1. The inequality constraints are

$$\begin{array}{rrcl} -x_1 & +x_2 & \leq & 2, \\ x_1 & +x_2 & \leq & 4, \\ -x_1 & & \leq & 0, \\ & -x_2 & \leq & 0. \end{array}$$

2. Each of the four constraints defines a half-plane, and the intersection of the half-planes gives the polyhedron in $\mathbb{R}^2$. We draw first the lines defined by the equality version of the constraints:

$$\begin{array}{rrcl} -x_1 & +x_2 & = & 2, \\ x_1 & +x_2 & = & 4, \\ -x_1 & & = & 0, \\ & -x_2 & = & 0. \end{array}$$

In order to identify on which side of the line is the half-plane, we use the gradient with the opposite sign. Indeed, the gradient of a function indicates the direction in which the function increases. Here, as the function must be less or equal to a constant, we need the direction in which the function decreases.

For instance, for the first constraint, the gradient is the first row of the matrix A. Written as a column vector, it is:

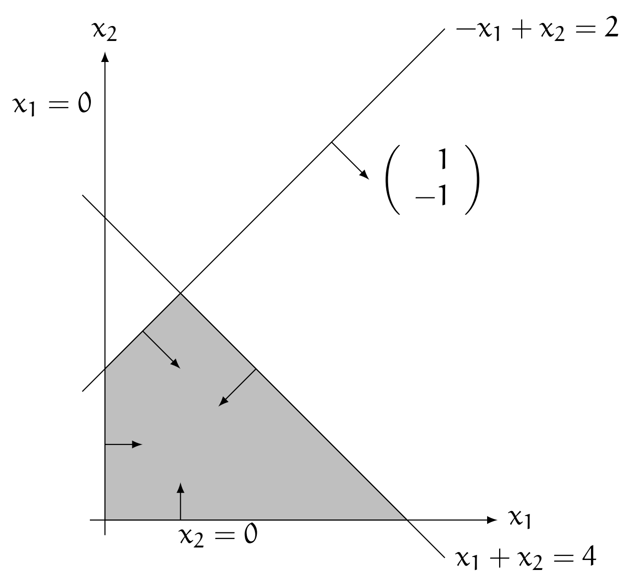
$$\begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Therefore, the direction that shows which side of the line is feasible is

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The other directions are obtained in the same way:

$$d_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},\ d_2 = \begin{pmatrix} -1 \\ -1 \end{pmatrix},\ d_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},\ d_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$



3. In order to write the standard form of the polyhedron, we first rewrite the inequality as follows:

$$\begin{array}{rrcl} -x_1 & +x_2 & \leq & 2, \\ x_1 & +x_2 & \leq & 4, \\ x_1 & & \geq & 0, \\ & x_2 & \geq & 0. \end{array}$$

   As required by the standard form, x_1 and x_2 must be non negative, so that nothing has to be done for that.

   Now, we need to include the slack variables. As there are 2 inequality constraints (different from the non-negativity constraints), we need 2 slack variables:

$$\mathcal{P} = \{x \in \mathbb{R}^2, x^s \in \mathbb{R}^2 \mid Ax + x^s = b, x, x^s \geq 0\}. \tag{1}$$

   We now have 4 variables. The set of constraints characterizing the polyhedron is:

$$\begin{array}{rrrrl} -x_1 & +x_2 & +x_3 & & = 2, \\ x_1 & +x_2 & & +x_4 & = 4, \\ x_1 & & & & \geq 0, \\ & x_2 & & & \geq 0, \\ & & x_3 & & \geq 0, \\ & & & x_4 & \geq 0. \end{array}$$